

# Basic Statistical Concepts

Moments · Estimators · MoM & MLE · Inference

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## Learning objectives

By the end of this chapter, you will be able to:

- interpret the *moments* of a distribution (mean, variance, skewness, kurtosis);
- compute classical *point estimators* (sample mean, variance, ACF) and discuss their properties (bias, consistency);
- apply two estimation methods: *Method of Moments* and *Maximum Likelihood*;
- describe the *sampling distribution* of standard estimators under normality;
- run a basic *statistical inference* procedure (confidence interval, hypothesis test,  $p$ -value).

## Chapter outline

1. Moments of a distribution
2. Classical estimators and sampling distributions
3. Estimation methods: MoM and MLE
4. Multivariate data
5. Statistical inference (CI, hypothesis testing)

## Moments of a distribution

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# 1. Raw and central moments

## Definition – Moments

Let  $X$  be a random variable with density  $f$  and mean  $\mu = \mathbb{E}[X]$ .

- **Raw  $\ell$ -th moment:**  $m'_\ell = \mathbb{E}[X^\ell] = \int x^\ell f(x) dx$ .
- **Central  $\ell$ -th moment:**  $m_\ell = \mathbb{E}[(X - \mu)^\ell]$ .

**Standard quantities:**

$$\mu = m'_1, \quad \sigma^2 = m_2 = \text{Var}[X], \quad S(X) = \frac{m_3}{\sigma^3}, \quad K(X) = \frac{m_4}{\sigma^4}.$$

## 1. Numerical example: $X \sim \mathcal{N}(2, 4)$

Theoretical moments:

$$\mu = 2, \quad \sigma^2 = 4, \quad S(X) = 0, \quad K(X) = 3.$$

Empirical estimates from  $n = 10\,000$  Monte-Carlo draws (seed 42):

$$\hat{\mu} \approx 2.005, \quad \hat{\sigma}^2 \approx 3.987, \quad \hat{S} \approx -0.013, \quad \hat{K} \approx 2.996.$$

*Skewness* measures asymmetry; *kurtosis* measures tail-heaviness. A normal distribution has  $S = 0$  and  $K = 3$  (excess kurtosis  $K - 3 = 0$ ).

## Classical estimators

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## 2. Sample mean and sample variance

Given i.i.d. data  $X_1, \dots, X_n$ :

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

### Proposition – Properties

Under i.i.d. assumption with finite second moments:

- **Unbiased:**  $\mathbb{E}[\bar{X}_n] = \mu$  and  $\mathbb{E}[S_n^2] = \sigma^2$ .
- **Consistent:**  $\bar{X}_n \xrightarrow{p} \mu$  and  $S_n^2 \xrightarrow{p} \sigma^2$  (Law of Large Numbers).



## 2. Sampling distribution under normality

If  $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ :

### Theorem

$$\bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{(n-1) S_n^2}{\sigma^2} \sim \chi_{n-1}^2, \quad \frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} \sim t_{n-1}.$$

- $\bar{X}_n$  and  $S_n^2$  are *independent* under normality (Cochran's theorem).
- These distributions are the basis of z-tests,  $t$ -tests and  $\chi^2$ -tests.

## 2. Central Limit Theorem

### Theorem – CLT

For i.i.d.  $X_1, \dots, X_n$  with  $\mathbb{E}[X_i] = \mu$  and  $\text{Var}[X_i] = \sigma^2 < \infty$ :

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Even when  $X_i$  is *not* normal, the sample mean is approximately normal for large  $n$ .  
Rule of thumb:  $n \geq 30$ .

## Estimation methods

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### 3. Method of Moments (MoM)

#### Definition – MoM

To estimate  $\theta \in \mathbb{R}^k$ , solve the system

$$\mathbb{E}_{\theta}[X^p] = \frac{1}{n} \sum_{i=1}^n X_i^p, \quad p = 1, \dots, k.$$

**Example:**  $\mathcal{N}(\mu, \sigma^2)$ .

$$\mu = \mathbb{E}[X] \Rightarrow \hat{\mu}_{\text{MoM}} = \bar{X}_n,$$

$$\sigma^2 = \mathbb{E}[X^2] - \mu^2 \Rightarrow \hat{\sigma}_{\text{MoM}}^2 = \frac{1}{n} \sum (X_i - \bar{X}_n)^2.$$

Note: this differs from  $S_n^2$  by a factor  $n/(n-1)$ .

### 3. Maximum Likelihood (MLE)

#### Definition – Likelihood and MLE

Likelihood:  $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$  for i.i.d. data. Log-likelihood:  $\ell(\theta) = \log L(\theta)$ . The MLE is

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \ell(\theta).$$

**Example:**  $\mathcal{N}(\mu, \sigma^2)$ . Solving  $\partial \ell / \partial \mu = 0$  and  $\partial \ell / \partial \sigma^2 = 0$ :

$$\hat{\mu}_{\text{MLE}} = \bar{X}_n, \quad \hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum (X_i - \bar{X}_n)^2.$$

Same as MoM for the Gaussian. Both are biased for  $\sigma^2$ .

### 3. Asymptotic normality of the MLE

#### Theorem

Under regularity conditions (identifiability, smoothness, finite Fisher information  $I(\theta_0)$ ):

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1}).$$

- MLE is **asymptotically efficient** (reaches the Cramér–Rao bound).
- MoM is simpler but generally less efficient.
- Both are consistent.

## Multivariate data

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## 4. Joint, marginal, conditional

Joint density:  $f(x, y)$ .

$$f_X(x) = \int f(x, y) dy, \quad f_{Y|X}(y | x) = \frac{f(x, y)}{f_X(x)},$$

$$f(x, y) = f_{Y|X}(y | x) f_X(x).$$

**Independence:**  $f(x, y) = f_X(x) f_Y(y)$ .

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y], \quad \text{Cov}(X, Y) = 0.$$



## 4. Sample covariance and correlation

Given paired data  $(X_i, Y_i)_{i=1}^n$ :

$$\widehat{\text{Cov}}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}),$$

$$\widehat{\text{Corr}}(X, Y) = \frac{\widehat{\text{Cov}}(X, Y)}{\hat{\sigma}_X \hat{\sigma}_Y} \in [-1, 1].$$

Correlation captures *linear* dependence only. Two variables can have  $\text{Corr}(X, Y) = 0$  and still be strongly dependent (e.g.  $Y = X^2$  with  $X$  symmetric).

# Statistical inference

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## 5. Confidence interval for the mean

Setting:  $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ .  $\sigma$  **known** (z-interval):

$$\text{CI}_{1-\alpha}(\mu) = \left[ \bar{X}_n - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right].$$

$\sigma$  **unknown** (t-interval):

$$\text{CI}_{1-\alpha}(\mu) = \left[ \bar{X}_n - t_{n-1, 1-\alpha/2} \frac{S_n}{\sqrt{n}}, \bar{X}_n + t_{n-1, 1-\alpha/2} \frac{S_n}{\sqrt{n}} \right].$$

**Reading:** probability  $1 - \alpha$  that the true  $\mu$  falls in the interval (over repeated samples).

## 5. Hypothesis testing

### Definition – Two-sided test on the mean

$H_0 : \mu = \mu_0$  vs.  $H_1 : \mu \neq \mu_0$ . Test statistic  $T = (\bar{X}_n - \mu_0)/(S_n/\sqrt{n}) \sim t_{n-1}$  under  $H_0$ . Reject  $H_0$  at level  $\alpha$  if  $|T| > t_{n-1, 1-\alpha/2}$ .

- **p-value:** probability under  $H_0$  of observing something at least as extreme as the data.
- **Type I error** ( $\alpha$ ): rejecting  $H_0$  when true.
- **Type II error** ( $\beta$ ): not rejecting  $H_0$  when false.
- **Power:**  $1 - \beta$  — sensitivity to detect  $H_1$ .

## 5. Numerical example

We sample  $n = 50$  Atlanta monthly temperatures (in  $^{\circ}\text{F}$ ). Sample statistics:  $\bar{X}_{50} = 61.2$ ,  $S_{50} = 16.8$ . Test  $H_0 : \mu = 60$  vs.  $H_1 : \mu \neq 60$  at  $\alpha = 0.05$ .

$$T = \frac{61.2 - 60}{16.8/\sqrt{50}} \approx 0.505, \quad t_{49,0.975} \approx 2.010.$$

Since  $|T| = 0.505 < 2.010$ , **do not reject**  $H_0$ .  $p$ -value  $\approx 0.616$ .

The 95% CI for  $\mu$  is  $[61.2 - 4.78, 61.2 + 4.78] = [56.4, 66.0]$ , which contains 60 — consistent with the test conclusion.

## Wrap-up

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## Key takeaways

- The *moments* characterise a distribution (mean, variance, skewness, kurtosis).
- *Classical estimators*  $(\bar{X}, S^2)$  are unbiased and consistent under i.i.d. assumptions.
- *MoM* and *MLE* coincide for the Gaussian; in general MLE is asymptotically efficient, MoM is simpler.
- Under normality,  $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$  and  $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$ .
- *CLT* extends asymptotic normality to non-Gaussian data.
- Confidence intervals and hypothesis tests are built on the sampling distribution of the estimator.

### **Chapter 2.** Time-series *decomposition*: trend and seasonality.

- Classical additive and multiplicative decomposition.
- Trend estimation by moving averages and parametric models.
- Seasonality estimation.
- Residual analysis.

**Thank you.**

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